

pLanck-99.

On-device malware classification for the devices nobody secures.

```
$ ./pLanck-99 --mode=inference --target=mcu
```

37 KB binary · no FPU · no OS · no network · a verdict in 34-62 ns

DIVISION-36

pre-seed · raising \$1-2M

investor deck · 2026

```
$ planck-99 --problem
```

13.5 billion devices run **blind**.

21.1B connected IoT devices shipped by end of 2025 – and 64% (≈13.5B) are MCU-class: no FPU, no OS, no network stack. Cloud and LLM defenses **physically** cannot reach them. Each is a silent entry point upstream of every grid and factory.

THE COST

Flax Typhoon (PRC-state) built Raptor Train from 260,000+ hijacked devices. One compromised IoT box is a foothold in critical infrastructure – and after Dec 2027, a board-level liability, not an engineering footnote.

- › 21.1B connected IoT devices (end-2025); **64% are MCU-class**, never securable – until now.
- › 260,000+ devices hijacked in a single nation-state botnet. The playbook is proven and repeatable.
- › Status quo = "do nothing," because nothing fits the hardware.

› blind spot • 13.5B devices • 64% unserved → now a board-level liability

```
$ planck-99 --why-now
```

A regulation just made the blind spot illegal.

EU CRA (2024/2847) forces named products to ship with on-device malware protection by a hard date. It is the first rule that requires MCU-class devices to detect malware where they run – exactly the tier nobody could defend.

COMPLIANCE DEADLINE

Dec 11, 2027

Non-compliance: **€15M or 2.5% of global turnover** – or removal from the EU market.

- > **EU CRA (2024/2847):** full obligations Dec 11 2027; reporting begins Sept 11 2026. No size exemption.
- > **Nation-state proof:** Volt Typhoon sat inside US critical infrastructure 5 years undetected – via unsecured embedded hardware.
- > **ICS vulnerabilities:** 2,155 CVEs in 2025 (508 advisories – first year >500); avg CVSS crossed 8.0.

> EU CRA 2024/2847 • hard deadline Dec 11 2027 → MCU detection now mandatory

```
$ planck-99 --gap
```

The defenders can't fit where the risk is.

Every existing approach assumes a CPU, an OS and a network. The risk lives precisely where those assumptions break.

PLANCK-99

Runs on bare metal – no FPU, no OS, no net.

37 KB. Fits inside the firmware build.

Deterministic; emits a CRA compliance pack.

EVERYONE ELSE

Cloud / LLM / EDR: need 50-500 MB RAM + FPU + OS + network. Embedded hardware gives 256 KB-4 MB.

Exein (~€700M, Linux+eBPF): stops at the OS boundary – can't reach no-FPU MCUs.

Status quo: "do nothing." Not a choice after 2027.

The gap is not a feature gap. It is a hardware boundary no one has crossed.

> incumbents need FPU+OS+net · we need none → the hardware boundary is crossed



```
$ planck-99 --solution
```

A syscall trace in. A verdict out. **Nothing else.**

Raw Linux syscall sequence in → a 5-tree pruned ensemble (INT8, integer-only C) → MALWARE / BENIGN + confidence. 29.4M inferences/sec on one core; 97.71% precision on unseen IoT malware. No OS calls at inference. No model download. No telemetry.

① TRACE

Capture the syscall sequence at the kernel boundary. The only input.



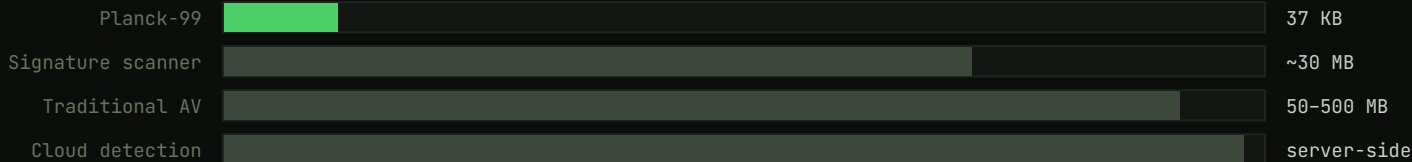
② CORE

32-dim normalized frequency vectors → INT8 closed-form dot product. 27 KB binary + 10 KB ring buffer. 34 ns. Deterministic.



③ VERDICT

MALWARE / BENIGN + confidence, plus a JSON proof file for CRA Art. 13.



Footprint vs existing approaches – Planck-99 (37 KB) is ~800× smaller than a signature scanner and ~13,000× smaller than traditional AV. It fits in L1 cache; the rest need MB of RAM or a data center.

> syscall trace → 5-tree INT8 ensemble → verdict • 37 KB • deterministic

```
$ planck-99 --integrate
```

OEMs pull it in. It doesn't push.

Distribution is the hard part of security hardware. Planck-99 removes the friction: it is the only detector that fits the bill of materials with zero dependencies.

- > Links into the existing firmware image – no new process, no RTOS requirement.
- > Compliance pack (CRA) is emitted at build time; auditors get evidence, not promises.
- > The constraint is the wedge: 37 KB + no deps is the only thing OEMs will accept onto a \$2 MCU.

WHY IT STICKS

Once Planck-99 is in the build, replacing it means re-certifying the whole device. Incumbents need years to reach this tier – by then the slot is taken.

> static .a + one call • CRA pack at build → OEMs pull it in

```
$ planck-99 --bench
```

Small model. Large margin.

98.88%

train / validation accuracy on
the pruned ensemble. Held-out
sets stay close.

92.01%

LinuxStatic held-out set

34-62 ns

inference (62 / 35 / 34 ns by
dataset)

96.28%

IoT Collected – 97.71%
precision

690×

less RAM than float32

1.45%

false-positive rate, balanced
set

15.8M×

faster than Python · ≤0.5 pp
lost vs float32

IoT FPR 12.18% is a distribution artifact (short benign traces, 1:5.3 ratio) – a 500-syscall gate mitigates it in production. OOD holds at 51×: trained on 2,275 events, accurate at 117,088.

> 98.88% acc · 34-62 ns · 690× RAM · 15.8M× speed · ≤0.5 pp vs float32



```
$ planck-99 --moat
```

Four properties incumbents can't retrofit.

Each one is a consequence of building for 37 KB — and a 1.4 MB peak-RAM ceiling — from the first line, not a feature bolted on later.

690×

less RAM than float32 — fits
MCUs nothing else fits

INT8

no FPU · no OS · no net — the
tier others can't reach

15.8M×

faster than Python — runs at
the kernel boundary

51×

OOD generalization — length-
invariant by construction

APPROACH	ON-DEVICE	AIR-GAPPED	REAL-TIME	MCU-CLASS
Traditional AV	○	○	●	○
Cloud detection	○	○	~	○
Signature scanner	~	●	○	○
OT platforms (Nozomi · Claroty)	○	~	~	○
Planck-99	●	●	●	●

Nearest peer **Exein** raised €170M in 2025 at ~€700M valuation — proof the category is real. But Exein needs Linux+eBPF and **cannot reach the 64% MCU tier** Planck-99 owns. Defensibility: a 5-tree INT8 ensemble only rebuildable at this size over years.

> 690× RAM · 15.8M× speed · INT8 · 51× OOD → 5-10 yr design lock-in

\$ `planck-99 --market` # bottom-up, not a analyst chart

\$0.20 × 100M devices.

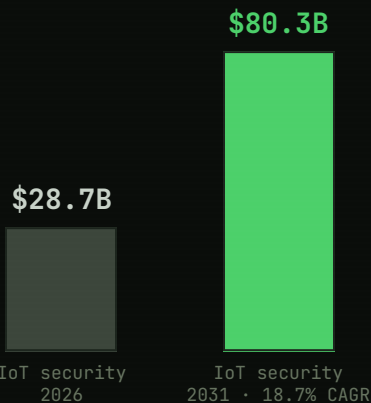
We size from the hardware we can actually ship to, not a top-down TAM.

13.5B devices unserved (64% of connected)

→ beachhead: 100M devices

× \$0.05-0.50 / device / yr

= \$5M-\$50M ARR at ~1% share



64% of 21.1B devices are MCU-class – our exact tier.
Europe alone: \$197.8B IoT market, first-priority under CRA.

Regulatory convergence is global, not just EU: UK PSTI (2024) · Japan JC-STAR (2024) · US NIST/FDA · Saudi NCA · UAE frameworks. On-device security is becoming a worldwide baseline.

> 100M devices × \$0.20 = \$20M ARR @1% · serviceable \$0.7-6.8B / yr

```
$ planck-99 --team
```

The bet at pre-seed is the founder.

Here it's a 17 y/o with a decade of experience – hacker, scientist, seller, all in one.

7

Linux
from docs

12

Marketing Ops
12-15 · off-sec

15

HackerOne
Top-90 · mentor

17

Planck-99
IEEE · Z-Jail

ZIAD SALAH · 17 · CAIRO, EG

Hacker – HackerOne #9 Egypt · Top-90 worldwide (VDP 2026); 12+ vulns, 7 critical in 48h. Axiom-WAF (99.33%) + SYRTH (4.1K rec/s) in production.

Scientist – IEEE AIITA 2026 **accepted** (PTRR); Nature (Scientific Reports) 4 review rounds, 9 reviewers. 15+ engines built solo.

Seller – Marketing specialist at 12-15: +75% client base in 6mo; declined Regional Mktg Manager to stay on research.

Z-Jail – native Linux sandbox featured in Risky Business (Jun 2026). Self-taught from age 7 · Certs: CRTP · CEH · eWAPTx.

Full record: github.com/Zierax/Ziad-Portfolio – every claim above is public & reproducible.

WHAT'S REAL TODAY

IP & filings in motion (patent pending). No legacy contracts – clean cap table by design.

Legal readiness: qualified guardian + IP counsel retained, operational today.

Business model: pure software SDK, ~100% gross margin – we make a lot and spend nothing to ship it.

Production tools live: Axiom-WAF, SYRTH, Planck-99 benchmarks – all public.

Honest gap: a COO with 15+ yrs deep-tech operating experience – the one seat left to build.

> bet on the founder: 17 y/o · hacker+scientist+seller · IP in motion · no legacy contracts

```
$ planck-99 --validation
```

Pre-revenue. Over-determined.

Deep-tech pre-seed doesn't show MRR. It shows proofs that compound. Each of these de-risks a different assumption.

- > **IP filings in motion** (patent pending); no contracts yet – clean cap table.
- > **Publications:** IEEE AIITA 2026 accepted; Nature (Scientific Reports) – 4 review rounds, 9 reviewers.
- > **Reproducible public benchmarks** – anyone can re-run the numbers.
- > **Legal readiness:** qualified guardian + IP counsel retained, operational today.
- > **Production tools shipping** – Axiom-WAF (99.33%), SYRTH (4.1K rec/s), Planck-99.
- > **Academic relations:** active collaborations across several top-tier universities.

> IP in motion • IEEE AIITA accepted • Nature 4 rounds • academic relations • reproducible

```
$ planck-99 --raise
```

Raising \$1-2M pre-seed.

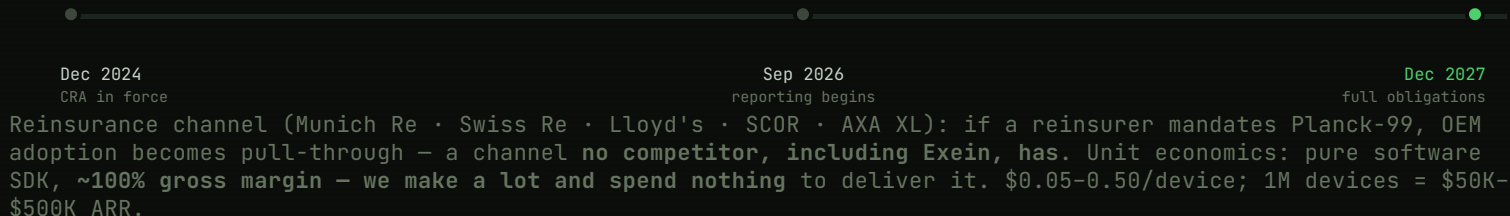
Tied to milestones that retire the core risks before the CRA window closes – not a department split.

MILESTONE 1 **Certify** – IEC 62443 / CRA conformance package. Unlocks OEM procurement.

MILESTONE 2 **2 lighthouse OEM pilots** – one defense, one industrial. Real deployment data.

MILESTONE 3 **Hire** commercial co-founder + 2 firmware engineers. The gap we named.

OUTCOME **Series A ready** on deployed evidence, before the Dec 2027 deadline. ~18-24 mo runway.



> \$1-2M → certify · 2 OEM pilots · hire co-founder → Series A ready



```
$ planck-99 --vision
```

A generational monopoly – and a life-mission to secure the world.

The EU CRA turns 13.5B previously unsecurable devices into a mandated market. Planck-99 is the only detector that fits the hardware. Whoever ships first becomes the default for a decade – and switching means re-certifying every device on the line.

THE MONOPOLY

A hard compliance mandate + a hardware boundary no incumbent has crossed = a closing window to become the standard. First to ship owns the slot; OEM lock-in does the rest. That is the generational return.

THE MISSION

More than a product: a decade-long commitment to secure the embedded world – from the \$2 MCU to the industrial grid. Built by someone who has lived both sides of the wire: the attacker and the scientist.

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> a generational monopoly · life-mission to secure the world · let's secure the 64%

